So, we have learned to make and visualize predictions from a prior GP without any training data. Now, let's actually train the GP model on the training set that we randomly generated and see how the predictions change. The nice thing about what we have coded so far is that everything may be repeated exactly, except for that we now initialize our GP with our training data (remember we used `None` for the first and second arguments before):

```
model = BaseGPModel(train_x, train_y, likelihood)
```

Rerunning the code will give us Figure 2.18.

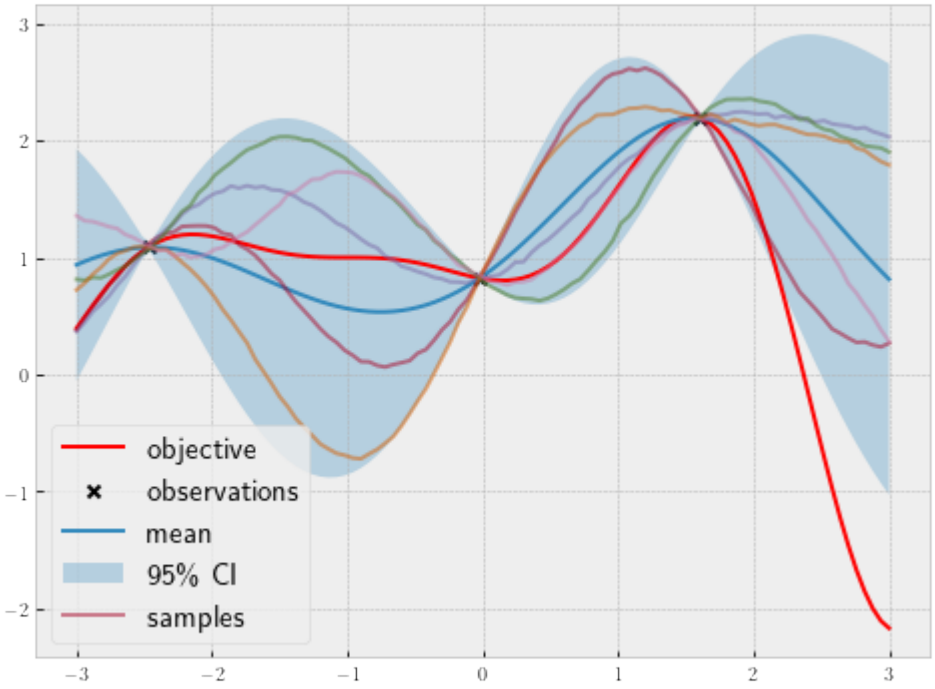


Figure 2.18 Predictions made by a posterior GP. The mean function and randomly drawn samples smoothly interpolate the training data points, while uncertainty vanishes in the regions surrounding these data points.

This is exactly the type of predictions that we'd like to see: the mean line and samples nicely interpolate our observed data points, and our uncertainty (measured by the CI) also reduces around those data points.

We can already see how this uncertainty quantification is useful in terms of modeling the objective function. Only after observing three data points, our GP has obtained a rather good approximation of the true objective function. In fact, almost all of the objective lies inside the 95% CI, indicating that our GP is successfully accounting for how the objective function might behave, even in regions where we don't have any data from the function yet. This calibrated quantification is especially beneficial when we actually need to make decisions based on our GP model, that is, when we decide at which points we should observe the function value to find the optimum, but let's save that for the next part of the book.

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